



Balancing Prediction and Constraint: An Optimization-Based Decision Support System for Paper Purchasing in Job-to-Order Print Manufacturing

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Executive Summary

This project tackles a typical operational decision in job-to-order print manufacturing: the amount of paper to buy for each production run, when the quoted material estimate for a job is not a reliable predictor of its actual consumption and is subject to real constraints on budget and storage. In support of this decision, an Optimization-Based Decision Support System was developed. The system combines a linear regression model that predicts order-specific paper cost based on business line, job type, and quantity, and a newsvendor-style optimizer that receives this prediction and a dynamically calculated monthly budget to produce a single recommended purchase quantity.

Validation against a batch of real historical orders showed that the system's recommendation was approximately 11% short of the actual material need under baseline assumptions – not a failure of the prediction or optimization logic, but because the organization's budget policy constrained the recommendation below the level that the underlying risk model would otherwise suggest. This was supported by structured scenario testing: tweaking the model's assumptions about risk didn't alter the final recommendation, but altering the budget policy did alter whether real need was met or grossly under-supplied. Therefore, the main recommendation of the project is that the organization needs to revisit the budget allocation policy, together with further model refinement, wider validation, and finally forecasting of incoming orders, so that the system's risk-aware purchasing logic can be used meaningfully in real purchasing decisions instead of being overridden by a fixed budget constraint.

Introduction

Print manufacturing is usually job-to-order. That means production doesn't start until a client confirms an order, and every job is different in size, paper type, and finishing requirements. This makes material planning much more difficult than in a factory that regularly produces the same products. There is no steady "recipe" that a print shop follows from month to month; there is only a quote that tells how much each job will take.

Print shops are always trying to figure out how much paper to purchase for jobs that are being printed and how much to keep on hand so that they do not experience production delays. If paper runs out in a press, there's lost production and delays in delivery. But on the other hand, buying too much ties up money in stock that might not suit future job needs.

The biggest problem is that the estimate for the quoted material never comes close to the actual amount used. Then there is the fixed amount of paper lost to press makeready before any usable output is produced, and variable amounts of waste depending on the particulars of the run, including misprints, adjustments, and job complexity. The analysis of the production data, both in quotes and completed orders, shows huge variations in quantities and material costs between job types, and this project is the bridge to that gap.

This is important because the decision to purchase involves two real costs: over-purchase and the cost of stopping production. This is generally based on the planner's experience and the quoted specification, with no structured approach to account for the fact that the actual material requirements of a job are often

different from the specification. This is the gap that the proposed decision support system wants to fill. It will use historical job data to produce a better estimate of paper requirements and then recommend a purchase volume to minimize costs and the risk of running out of stock, taking into account storage and budget limitations.

Problem Definition

At the start of each production cycle, the organization has to decide how much paper to buy for production jobs. This is a difficult decision because the quoted material estimate for a job is not a reliable predictor of the actual paper usage. This uncertainty is caused by fixed press makeready waste as well as other factors such as job complexity and business line. The organization faces two conflicting risks in this decision: buying too little paper, which stops production and delays delivery, and buying too much paper, which uses up capital and storage space without meeting current production needs.

The Purchasing or Production Manager is the decision maker. This person approves the purchase of materials for each cycle. It's a balancing act of material availability, cost efficiency, and the consequences of overbuying or underbuying.

Stakeholders

- *Production staff*: There must be enough materials to complete jobs on time.
- *Finance*: They worry about capital locked up in inventory and staying within budget constraints.
- *Sales and client-facing staff* – Their commitments to clients are dependent on uninterrupted production. Material shortages compromise this.
- *Warehouse/storage personnel*: Their constraint is the physical space to store purchased materials.

Objectives

- To minimize the total expected costs, balance the holding costs of unused paper with the costs from production downtime due to material shortages. This insight was followed by:
- Close the gap between quoted estimates of material and actual paper usage by basing purchase recommendations on historical order data instead of only quotes.
- Ensure that the suggested purchase takes into account actual demand uncertainty, not just a single expected number, thus minimizing the possibility of a production halt.

Constraints

- A monthly limit on the percentage of the total value of orders in production during that cycle. This helps to ensure that spending on materials moves with business activity.
- A fixed physical storage capacity, i.e., the maximum amount of paper stock that can be stored at a time.

DSS Architecture

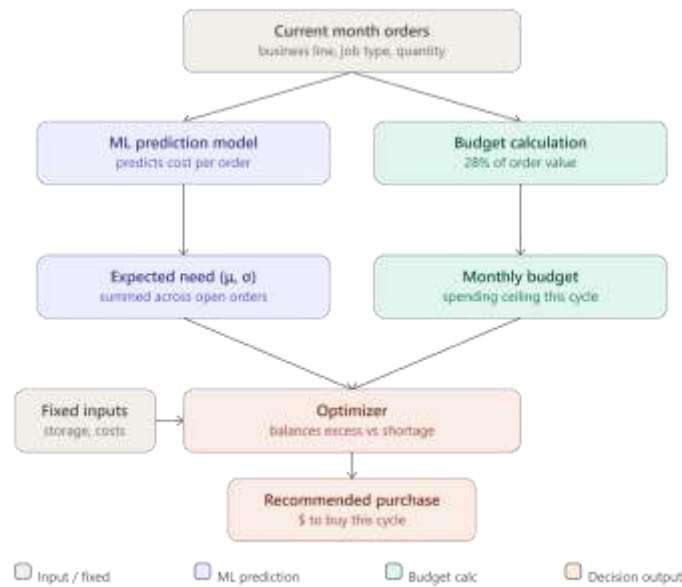


Figure 1: DSS Architecture for the Paper Purchase Optimization System

The diagram outlines the journey of a single decision, starting from raw data and leading to a final recommendation as it processes through the system in each production cycle.

This month's open orders are the start of the story. These are the jobs the shop has committed to make, and each one has a business line, a type of job, and a quantity. From this initial point, the same data is processed in two ways. Two different types of information are needed before making a purchase decision.

The ML prediction model considers patterns from one year of historical production data. It looks at the actual paper consumed by similar past orders (grouped by business line and job type) once produced. This helps estimate a realistic material need for each of this month's orders, correcting for the difference between what a job was quoted and what it usually consumes in practice. These order-level estimates are combined into an expected total need (μ) and a measure of the uncertainty of that need (σ). Meanwhile, the budget calculation uses a fixed business rule on the same orders to establish a spending limit based directly on their total value.

Both outputs are key inputs to the optimizer, the system's decision-making hub. The optimizer accounts for the predicted need and uncertainty from the prediction model and the spending limit from the budget calculation. It also considers the physical storage limit and the organization's actual cost structure (balancing the cost of holding idle paper and the cost of running out) to determine the purchase quantity that minimizes total expected cost. Without the ML model's output, the optimizer would lack a realistic estimate of what to buy. Without the budget calculation's output, it would not have a realistic affordability limit. There is no single pathway that will produce a usable decision. The recommendation only emerges once the optimizer combines both inputs.

The final output—the recommended purchase—is the result of the optimizer synthesizing a learned estimate of reality from the ML model with a business constraint from the budget rule into one clear, actionable figure.

DSS Methodology

Selected approach: Optimization-Based DSS

The project is based on an optimization-based Decision Support System. This method was chosen because the main decision of how much paper to buy each cycle has key features of an optimization problem. It has a clear goal: to minimize total expected cost. It also involves a clear decision variable, the purchase quantity, and specific limits like budget and storage capacity. We considered other options but discarded them. A Group Decision Support System was not relevant as the decision does not need to reconcile different stakeholder views or reach a consensus. A Simulation-Based Decision Support System was not chosen because the problem does not involve modeling a changing process over time. Instead, it focuses on a single-period purchase decision made under uncertainty.

Specific technique: newsvendor model

The optimization category presented the decision as a newsvendor problem. This procedure is suitable for situations where a single purchase quantity has to be determined before the uncertain demand is known. The cost of overbuying is not the same as the cost of underbuying. This is perfect for the paper buying decision. There are costs in holding excess paper, such as tied-up capital and storage.

On the flip side, not buying enough leads to a much higher cost due to production downtime. The newsvendor model explicitly addresses this imbalance. It is the amount of purchase above the expected need as a function of both the degree of uncertainty and the severity of a shortage relative to an excess.

Role of the prediction model

To work, the newsvendor model needs two inputs: an estimate of expected demand and a measure of its uncertainty. A linear regression model is used to create an order-specific estimate instead of using a simple historical average for these inputs. This estimate is based on the business line and job type of orders in production for that cycle. This prediction step is a supporting input to the optimization model rather than an independent decision support system. The decision support system is still optimization. The regression model is just there to give the optimizer a better estimate of expected need than a static average would give.

System Design & Implementation

Overview:

This system picks up the factory's open orders, computes the paper requirements, estimates the available cash, and gives a purchase amount that balances the risk of shortages and overspending.

The three main components are:

1. **The prediction:** It takes each order (business line, job type, quantity) and estimates the paper needed based on what similar orders have actually used in the past.
2. **The budget:** It looks at the same orders and calculates how much money can be spent this month, based on the total orders for the month.
3. **The optimizer:** It computes the actual amount to be purchased based on the prediction estimate and the money limit from the budget, taking into account the risk of running out and the risk of overbuying. It examines each order (business line, job type, quantity) and estimates how much paper is needed based on the actual usage of similar past orders.

Technical Description:

Category	Detail
Data source 1	<i>master_cotizaciones.csv</i> ("cotizaciones" = quotes) — sole source of <i>insumos_total</i> ("insumos" = supplies/materials), used to build the prediction target
Data source 2	<i>master_ordenes_ejecutivo.csv</i> ("ordenes" = orders) — authoritative source, reflects real, completed production orders
Join method	Matched by order number; retained only quotes that were both quoted and actually produced
Excluded variable	<i>repuestos_valor</i> ("repuestos" = spare parts, "valor" = value) — confirmed to be machine spare parts cost, unrelated to paper consumption
Business line categories	EDITORIAL (publishing/book-line business) and PUBLICOMERCIAL (commercial/advertising-print business); EMPAQUE (packaging) and CAJAS PASTA DURA (hardcover boxes/cases) folded into PUBLICOMERCIAL due to insufficient sample size (19 and 2 orders respectively)
Job type derivation	Keyword-based classification of the trabajo (" <i>trabajo</i> " = job/work) description text (e.g., BOOK/LIBRO, FLYER/VOLANTE, CATALOG/CATÁLOGO, AGENDA/CUADERNO)
Prediction model type	Linear regression (selected for coefficient interpretability over less transparent alternatives)
Model features	Business line (<i>linea_negocio</i> = business line), job type, <i>cantidad</i> (= quantity)
Model target	Total paper cost per order (<i>insumos_total</i> × 70%, reflecting paper's estimated share of total material cost)
Data split	75% training / 20% testing / 5% held-out validation batch
Optimization model type	Newsvendor model (single-period purchase decision under uncertain demand with asymmetric over/under costs)
Optimizer inputs	Expected need and uncertainty (from prediction model), monthly budget, storage capacity, holding cost rate, stockout cost rate
Optimizer output	Recommended paper purchase quantity (\$) for the current cycle

Tools/libraries	Python; pandas (data handling); scikit-learn (regression); scipy (optimization statistics)
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Rules and thresholds of the model

Rule 1 - Prediction Model: The system uses a trained linear regression to predict the expected cost of paper for an order. It is stated as:

$$\text{Predicted paper cost} = \text{Baseline} + \text{Business line adjustment} + \text{Job type adjustment} + \text{Quantity} * \text{Rate}$$

The trained coefficients correspond to a baseline of \$695.74. This is adjusted by +\$174.29 for orders of type EDITORIAL or -\$174.29 for orders of type PUBLICOMERCIAL. There is also an additional adjustment of +\$141.30 for BOOK type jobs, and smaller adjustments for other job types. The quantity effect is small at about \$0.002/unit.

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Baseline (intercept): $695.74
business_line_EDITORIAL      +174.2901
business_line_PUBLICOMERCIAL -174.2901
job_type_BOOK                 +141.2966
job_type_FLYER                -41.8574
job_type_POSTER_DISPLAY       -29.3987
job_type_LABEL                -23.7149
job_type_BROCHURE             -22.8033
job_type_CARD                 -18.5476
job_type_MAGAZINE             -15.5938
job_type_CATALOG              +14.9158
job_type_OTHER                +10.5481
job_type_STICKER              -10.5450
job_type_AGENDA               +9.6589
job_type_BOX                  -8.6861
job_type_FOLDER               -3.3761
job_type_NEWSPAPER            -2.9471
job_type_CALENDAR             +1.0506
cantidad                      +0.0023
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Figure 2: Coefficients of Linear Model Variables

Rule 2 - Budget calculation: The system calculates each cycle's spending limit in the following way:

$$\text{Monthly budget} = 28\% * (\text{total amount of the production orders of that cycle})$$

The 28% figure is the material cost specific to paper as a proportion of order value. (Forty percent of order value = material-related cost; 70 percent of that = paper)

Dynamic budget (28% of order value): \$44,950.20
 Storage capacity: \$300,000.00
 Ideal purchase (unconstrained): \$59,315.92
 RECOMMENDED PURCHASE THIS CYCLE: \$44,950.20
 Binding constraint: budget

Figure 3: Constraints on Purchase Capacity

Rule 3. Optimization (newsvendor) rule: The system calculates the optimal purchase quantity by adding a safety margin to the expected demand.

$$\text{Ideal Purchase} = \text{Expected need } (\mu) + (\text{Uncertainty } (\sigma) * \text{Safety factor})$$

The safety factor is derived from the relative cost of understocking and overstocking, which is known as the critical ratio.

$$\text{Critical ratio} = \text{Stockout cost rate} / (\text{Holding cost rate} + \text{Stockout cost rate})$$

This ratio represents the purchase level at which the expected cost of buying one more unit of paper (if not used) equals the expected benefit of that unit (if it prevents a shortage). The normal distribution transforms the resulting ratio into a safety factor. This point in the demand distribution is where the organization is protected against understocking at the desired confidence level. The last recommended purchase is the best purchase that fits in the budget or storage space, whichever is smaller.

Recommended purchase: \$44,950.20
 What was actually needed: \$50,484.10
 Gap: \$-5,533.89 (-11.0%) (would have fallen short)

Figure 4: Optimizer recommendation of 5% batch OP reserve for testing.

Rule 4 — Cost assumptions.

Assumption	Value	Basis
Holding cost rate	1.5% (tested up to 5%)	Reflects the organization's buying on credit terms and keeps the cost of holding extra material low.

Stockout cost rate	0.408	Derived from the organization's estimated idle-time cost of \$500 per hour, the share of that cost related to materials and paper, and normal operating hours of 176 hours per month.
Storage capacity	\$300,000	Fixed physical limit on paper stock.

Process Flow:

Each production cycle of the system goes through a fixed sequence of six steps:

Step 1. Enter current orders: The user enters a list of open production orders for the cycle, including for each order the business line, job type, quantity, and quoted value.

Step 2. Predict paper requirements per order. The trained regression model considers the characteristics of each order to produce an individual predicted paper cost.

Step 3. Aggregate Expected Need and Uncertainty: The individual forecasts for each order are added together to give a total expected requirement, (μ), for the cycle. The associated uncertainty, σ , is then estimated based on the model's known prediction error.

Step 4. Work out the budget: At the same time as Steps 2 and 3, the total quoted value from the same order list is used to calculate the spending ceiling for the cycle (28% of the total order value).

Step 5. Execute the optimizer: Expected demand, uncertainty, and budget are inputs to the newsvendor optimization model. It also assumes a fixed storage capacity and cost rate. This model will determine the optimal quantity to purchase within budget and storage constraints.

Step 6. Print the recommendation: The system will return a single recommended purchase amount and will indicate the constraint (if any) that limited the recommendation. So far, the budget constraint has always been the limiting factor in testing.

Evaluation & Scenario Analysis

The strength of the purchase recommendation was tested by evaluating the model in four scenarios. The base case is the current best-estimate operating conditions. Each of the three alternative scenarios changes one set of assumptions to show a different operational or stakeholder perspective. The worst-case scenario

accounts for the effect of higher operational risk. The shop is operating at full capacity. The finance-conservative and production-conservative scenarios consider the effect of tightening or loosening budget policy. These scenarios stem from the different priorities of finance, concerned with capital efficiency, and production, which prioritizes a safety margin against shortages, as stakeholders in the purchasing decision. In each scenario, the same batch of 55 real historical orders was used for model validation. This way, you can directly compare the recommended purchase with the actual cost of paper for that batch.

Thresholds Scenarios

Scenario	Stockout cost rate	Holding cost rate	Budget factor	What changes vs. base
Base case	0.408	1.5%	28%	— (reference point)
Worst case (operational stress)	1.020	1.5%	28%	Stockout cost rate raised, reflecting the shop running at maximum 20hr/day capacity instead of a normal 8hr shift
Finance-conservative	0.408	5%	20%	Holding cost raised (finance's more conservative view of capital cost); budget factor tightened.
Production-conservative	0.408	1.5%	35%	Budget factor loosened (production's priority: safety margin over capital efficiency)

Results Comparison:

Scenario	Budget	Ideal purchase (unconstrained)	Recommended purchase	Binding constraint	vs. actual need (\$50,484)
Base case	\$44,950	\$59,316	\$44,950	Budget	–11.0% (short)
Worst case (operational stress)	\$44,950	\$62,878	\$44,950	Budget	–11.0% (short)
Finance-conservative	\$32,107	\$53,893	\$32,107	Budget	–36.4% (short)

Production-conservative	\$56,188	\$59,316	\$56,188	Budget	+11.3% (covered)
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In all four cases, the recommendation was risk-based rather than budget-based. Simulating operational stress (worst case) by increasing the stockout cost rate raised the ideal purchase without constraints from \$59,316 to \$62,878. The actual recommendation, however, remained at \$44,950. Only two scenarios that changed the budget policy affected the outcome. The budget was tightened under finance-conservative assumptions, and the recommendation was reduced to \$32,107, or 36.4% short of actual need. Conversely, relaxing the budget under production-conservative assumptions raised the recommendation to \$56,188. It was the sole scenario to satisfy actual need, by an 11.3% margin.

Critical Discussion

Strengths

The system also uses a machine learning model that learns the specific material needs of each order, rather than a single fixed estimate for all jobs. This allows the predicted paper requirement to be derived from the actual attributes of each batch of orders and, consequently, a purchase recommendation to be derived based on the actual order mix of the current month, not just a generic historical average.

Furthermore, the system's cost assumptions (holding cost rate, stockout cost rate, budget factor) are based on the organization's operational data. Still, they can be changed, allowing the model to support structured scenario testing under different business conditions.

Limitations

The explanatory power of the prediction model is still small ($R^2 = 0.11$), meaning that the current feature set (business line, job type, and quantity) does not explain most of the variance in paper cost per order.

The full pipeline was validated on a single held-out batch of historic orders. Validation on multiple, and especially larger, batches would be needed—particularly months with higher production volume and more open production orders—to confirm that the system's performance generalizes beyond the specific batch on which it was tested.

In addition, to function effectively as an ongoing decision support tool, the system would ultimately require the capability to estimate incoming production orders for the month after the purchasing decision, not just the orders that were on hand at the time the decision was made.

Another possible source of bias risk is a new production order that is unlike any of the job types the prediction model has been exposed to before. Then the model has no historical basis to estimate the material needs of such an order, and one cannot assume that the prediction accuracy for truly new types of jobs will match the accuracy it has for well-represented categories.

Practical applicability

The system is intended to work directly with the data the organization is already collecting from its existing quotation and production order records, so no new data collection infrastructure is required to execute it.

In practice, two further developments would help adoption: * embedding a forecasting capability for incoming material purchases into the optimizer itself, instead of prediction and optimization being separate steps * formally combining prediction and budgeting into one coherent workflow.

Decision-quality improvements

The system replaces an informal, experience-based purchasing decision with a structured, data-based, and risk-aware recommendation. In addition to providing a single recommended figure, the scenario testing performed in this project revealed a decision quality insight that likely would not have been quantified with an unstructured approach: that the organization's current budget policy is the dominant factor constraining the purchase recommendation. This result demonstrates that the value of the DSS is not just in providing a single answer, but in exposing the business assumptions that most influence the decision.

Ethical & Organizational Considerations

Fairness: The pattern of paper consumption in dollars for every order type is learned from historical data. Each production order (OP) is considered a separate case and compared with similar past orders, instead of using one estimated value for all jobs. This means that the recommendation can be made based on real features of the order being analyzed rather than treating dissimilar jobs uniformly.

Transparency: The prediction model and the optimizer were intentionally kept simple (i.e., a linear regression and a newsvendor optimization model, respectively) to keep the outputs traceable and auditable, instead of a black box. Any organization member, irrespective of technical background, can be shown why a specific recommendation was generated: which business line or job type adjustment led to the predicted cost, and which constraint (budget or storage) influenced the ultimate purchase number.

Accountability: The system is providing a recommendation, not an automatic purchasing action. The logistics or production manager takes the final decision to approve a material purchase. The manager is responsible for the final decision and can compare the system's output with the operational context the model is not able to capture.

User acceptance: The system is designed to complement rather than replace users' judgment. This allows users to directly interact with the assumptions that drive a recommendation and understand the reasoning behind it, rather than having to trust an opaque output, as the underlying constraints are adjustable and the model's outputs can be inspected across different scenario levels.

Potential biases: The optimizer's suggestions are structurally biased to prevent understocking, as the stockout cost rate is much larger than the holding cost rate in the current cost structure. This bias reflects the asymmetry of the organization's costs.

Conclusion & Recommendations

This project developed an Optimization-Based Decision Support System to address a real recurring decision in job-to-order print manufacturing: how much paper to purchase each production cycle under uncertainty about actual material consumption and real constraints on budget and storage. The system combines the two pieces into one working pipeline – a linear regression model that predicts paper cost per order based on business line, job type, and quantity, and a newsvendor-style optimizer that takes that prediction and a dynamically calculated budget and converts it into a single recommended purchase quantity.

The pipeline is end-to-end and provides a justifiable and explainable recommendation validated on a held-out batch of real historical orders. However, this validation also produced an important finding: under baseline assumptions, the recommended purchase was approximately 11% less than the batch's actual material need because the organization's budget policy constrained the recommendation below what the risk-based optimization would otherwise have suggested. This finding was further reinforced by structured scenario testing, which demonstrated that the model's risk assumptions (cost rates for stockouts and holding costs) did not influence the final recommendation. Still, the budget policy did directly determine whether the recommendation was appropriate to real need or fell far short of it. This suggests that in its present form budget policy is the most important factor in determining purchasing outcomes.

Recommendations:

Review the budget policy. Scenario testing identified the budget constraint as the primary factor preventing the recommendation from meeting real material need, and the organization should consider whether the current budget allocation represents a conscious risk tolerance or an unexamined default.

Add order forecast. Today, the system only responds to production orders that are already available. The model could be extended to estimate orders that are expected to enter the pipeline mid-cycle, allowing the purchase decision to be based on expected future need, rather than on current demand.

Extend validation over multiple batches. Testing was done on one held-out batch of historical orders. Validation of the system against other batches, particularly higher-volume months, would enhance confidence that its performance generalizes beyond the specific case analyzed here.

Increase predictive model accuracy. The present model explains a small fraction of the variation in paper price ($R^2 = 0.11$). Future iterations may include additional order-level features, such as page count or paper specification, if that data becomes available, to improve predictive accuracy while maintaining the model's current transparency.

Align the budgeting and prediction processes. The prediction and the budget calculation currently run as two separate, but coordinated processes. By combining these in a common workflow, you could reduce

the need for manual coordination and help foster more consistent, repeatable use of the system in ongoing operations.

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AI declaration

Claude (Anthropic, 2025) was an artificial intelligence tool employed to support text editing, add humanism to writing style, and assist in developing analytical storylines. The authors make all decisions regarding analysis, interpretation, and conclusion.